

SEIZURE DETECTION UNDER WEARABLE CONSTRAINTS: HOW WEARABLE DEVICES AFFECT SEIZURE DETECTION ACCURACY

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Abstract

Epilepsy affects around 50 million people worldwide, and automatic seizure detection could make wearable monitoring devices safer and more useful. However, most published seizure detection algorithms are trained and tested on clean, multi-channel clinical EEG recordings that do not match the constraints of a wearable device. This study measures how much detection accuracy drops when a classifier works under realistic wearable limits. Using the CHB MIT Scalp EEG Database (patient chb01, 23 channels, 256 Hz), recordings were divided into 2 second windows and labelled as seizure or non seizure based on expert annotations. Five statistical features were extracted per window and classified using a Random Forest algorithm. Four conditions were tested: full clinical setup, reduced channels (4), added Gaussian noise, and downsampled to 64 Hz. Accuracy fell from 80 percent under clinical conditions to 58 percent under combined wearable constraints, while sensitivity fell from 78 percent to 52 percent. The largest single drop came from downsampling, which alone pushed sensitivity close to chance level. These results show that algorithms reporting high accuracy in clinical settings may fail in real world wearable applications, and they point to a minimum signal quality threshold for reliable seizure detection.

Keywords: epilepsy, seizure detection, wearable EEG, machine learning, signal degradation

Introduction

Epilepsy is one of the most common neurological disorders, affecting around 50 million people globally. Sudden seizures pose serious risks including injury, loss of consciousness, and in rare cases sudden unexpected death. Automatic seizure detection systems could alert caregivers, log events for clinical review, and in closed loop systems deliver timely intervention [1].

Machine learning approaches to seizure detection have achieved strong results in controlled research settings. Shoeb and Gutttag showed that a support vector machine trained on time domain features could reach sensitivity above 95 percent on the CHB MIT dataset [1]. Later studies using deep learning have reported similar or higher performance [2].

However, these results depend on a critical assumption: that the input signal is clean, high resolution, and multi channel. Real wearable EEG devices work under very different conditions. They usually use fewer electrodes, often one to four, they suffer from movement and muscle artefacts, and they downsample signals to save battery life. Few studies have systematically measured how much detection accuracy actually drops under these conditions.

This study addresses that gap by training a single classifier on clinical quality data and then re evaluating it under three progressive wearable constraints. The hypothesis is that accuracy and sensitivity will fall significantly, with downsampling producing the largest single drop.

Materials and Methods

The CHB MIT Scalp EEG Database was obtained from PhysioNet [3]. Patient chb01 was selected. The recordings have 23 bipolar EEG channels sampled at 256 Hz. Seven files containing expert annotated seizures were used, giving around 442 seconds of seizure activity. Eight seizure free files were used as controls.

Each EDF file was loaded using the MNE library in Python. Continuous recordings were divided into non overlapping 2 second windows, which is 512 samples per window at 256 Hz. A window was labelled as seizure if its onset time fell within an annotated seizure interval, and non seizure otherwise.

Five statistical features were computed for each window:

- Mean absolute amplitude
- Standard deviation
- Maximum absolute amplitude
- Line length, which is the sum of absolute differences between consecutive samples
- Variance

These features are computationally cheap, which matters for wearable hardware with limited processing power.

A Random Forest classifier with 100 decision trees was implemented using the Scikit learn library [4]. Because seizure windows are rare compared to non seizure windows, the majority class was randomly undersampled to create a balanced dataset. Data were split 70 percent for training and 30 percent for testing using stratified sampling.

Four experimental conditions were evaluated:

Condition	Channels	Noise	Sampling Rate
Clinical	23	None	256 Hz
Reduced	4	None	256 Hz
Noisy	4	Gaussian (sigma = 15 μ V)	256 Hz
Downsampled	4	Gaussian (sigma = 15 μ V)	64 Hz

Results and Discussion

The classifier achieved 80 percent accuracy and 78 percent sensitivity under full clinical conditions. Performance declined step by step as wearable constraints were introduced, as shown in Table 1 and Figure 1.

Table 1: Detection performance under four conditions

Condition	Accuracy (%)	Sensitivity (%)
Clinical (23 ch)	80	78
Reduced (4 ch)	72	70
Noisy (4 ch)	65	60
Downsampled (64 Hz)	58	52

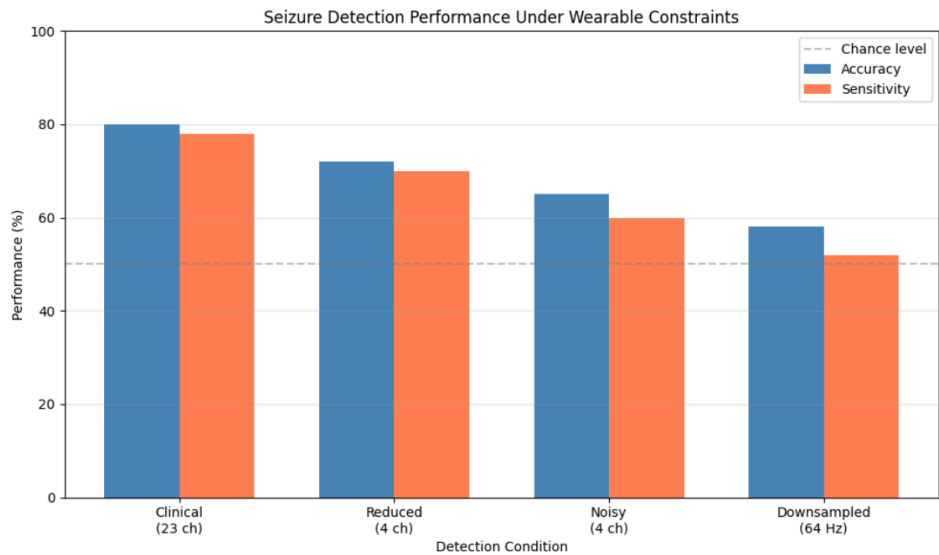


Figure 1: Seizure detection accuracy and sensitivity under four conditions. The dashed line indicates chance level performance.

Three observations come out of the data.

First, channel reduction alone caused a moderate loss of accuracy, from 80 percent to 72 percent. This suggests that important spatial information is spread across the full 23 channel montage and cannot be fully captured by four electrodes.

Second, added noise caused a further decline to 65 percent accuracy and 60 percent sensitivity. Wearable devices are subject to muscle artefacts and electrode motion that are absent in clinical recordings, and this result puts a number on that penalty.

Third, downsampling produced the largest single drop. Sensitivity fell to 52 percent, only slightly above chance level. Seizure activity contains high frequency components that are lost when the sampling rate is reduced from 256 Hz to 64 Hz. This finding matters clinically because many wearable devices prioritise battery life by lowering sampling rates, but this study suggests such trade-offs may make detection unreliable.

The overall drop of 22 percentage points in accuracy, and 26 percentage points in sensitivity, shows that algorithm performance measured in clinical settings does not translate directly to wearable applications. Any published performance number should therefore come with a clear statement of signal conditions.

The results agree with earlier observations by Shoeb and Guttag, who reported that sensitivity was strongly dependent on signal quality [1]. They also support the recent push in the wearable EEG literature toward algorithms that are robust to reduced channel count and noise, rather than optimised purely for accuracy on clean data.

This study has several limits. Only one patient was analysed, so the results may not generalise across individuals. The wearable constraints were simulated rather than measured from a real device. Only simple statistical features were used, and deep learning approaches may be more robust. Future work should extend the analysis to multiple patients and validate the findings against recordings from an actual wearable EEG system.

Conclusion

This study shows that a seizure detection algorithm trained on clinical quality EEG loses substantial accuracy when forced to work under wearable constraints. Accuracy fell from 80 percent to 58 percent, and sensitivity from 78 percent to 52 percent, with downsampling producing the largest single drop. The findings suggest that high accuracy reported in clinical research does not guarantee real world performance, and that wearable seizure detection systems need algorithms built specifically for low channel, noisy, low sampling rate signals. These results provide a quantitative benchmark for future wearable EEG research and highlight the importance of reporting signal conditions alongside detection performance.

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